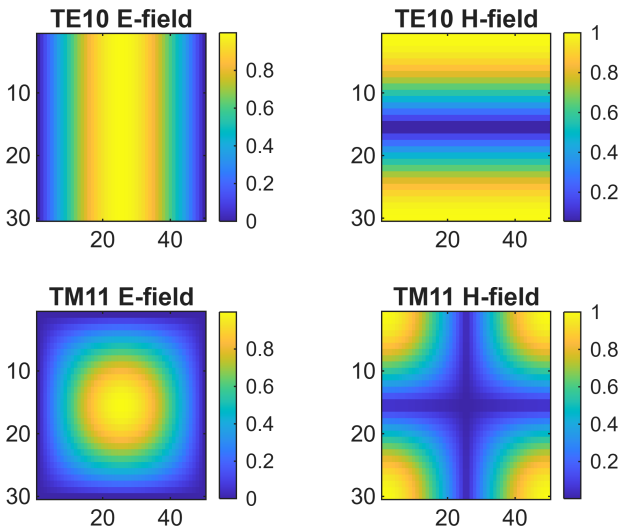
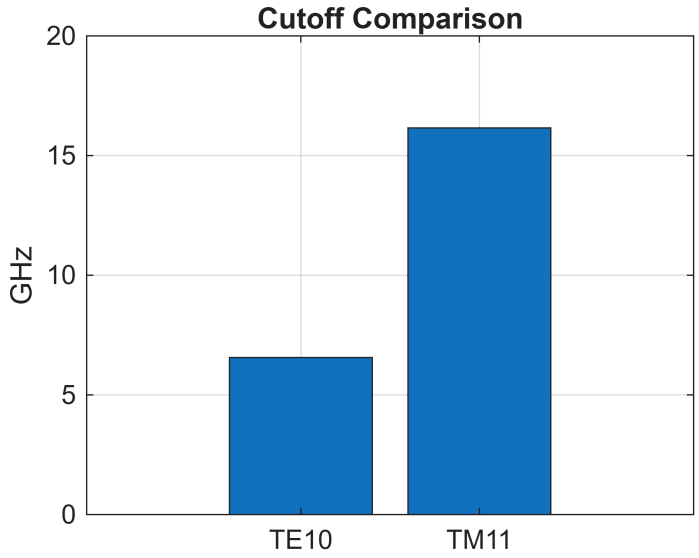
```
clc; clear; close all;
c=3e8; a=22.86e-3; b=10.16e-3; f=10e9;
fc10=c/(2*a); fc11=c/2*sqrt((1/a)^2+(1/b)^2);
beta=2*pi*f/c*sqrt(1-(fc10/f)^2);
fprintf('TE10 Cutoff=%.3f GHz\nTM11 Cutoff=%.3f GHz\nTE10 Beta=%.2f rad/
m\n',fc10/1e9,fc11/1e9,beta);
```

```
TE10 Cutoff=6.562 GHz
TM11 Cutoff=16.156 GHz
TE10 Beta=158.05 rad/m
```

```
x=linspace(0,a,50); y=linspace(0,b,30); [X,Y]=meshgrid(x,y);
figure;
subplot(2,2,1); imagesc(abs(sin(pi*X/a))); title('TE10 E-field'); colorbar;
subplot(2,2,2); imagesc(abs(cos(pi*Y/b))); title('TE10 H-field'); colorbar;
subplot(2,2,3); imagesc(abs(sin(pi*X/a).*sin(pi*Y/b))); title('TM11 E-
field'); colorbar;
subplot(2,2,4); imagesc(abs(cos(pi*X/a).*cos(pi*Y/b))); title('TM11 H-
field'); colorbar;
```




```
figure; bar([fc10 fc11]/1e9); set(gca,'XTickLabel',{'TE10','TM11'});
ylabel('GHz'); title('Cutoff Comparison'); grid on;
```




```
F=7:0.1:20; B=(2*pi*F*1e9/c).*sqrt(max(0,1-(fc10./(F*1e9)).^2));
figure; plot(F,B,'b','LineWidth',2); hold on; yline(0,'r--');
xlabel('Frequency (GHz)'); ylabel('\beta (rad/m)'); title('TE10
Propagation'); grid on;
```

